

Instantaneous Magnetic Field Distribution in Brushless Permanent Magnet dc Motors, Part I: Open-Circuit Field

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Abstract—An analytical technique for predicting the instantaneous magnetic field distribution in the airgap region of radial-field topologies of brushless permanent magnet dc motors, under any specified load condition and accounting implicitly for the stator winding current waveform and the effect of stator slot-openings, has been developed. It is based on the superposition of the component fields due to the permanent magnet and the stator excitation. In this paper, which is one of a series of four, a 2-d analytical method for predicting the open-circuit airgap field distribution in both internal and external rotor radial-field motor topologies is presented. It involves the solution of the governing field equations in polar coordinates in the annular airgap/magnet region of a multi-pole slotless motor in which the magnets are assumed to have uniform radial magnetization and a constant relative recoil permeability. Results for various radial-field motors are compared with predictions from corresponding finite element analyses.

I. INTRODUCTION

A GENERAL analytical technique for predicting the open-circuit magnetic field distribution in radial-field topologies of brushless permanent magnet motors is a prerequisite for force calculations, and hence the prediction of acoustic noise and vibration spectra [1]; for the prediction of the back-emf waveform—for steady-state and transient dynamic performance simulations [2] and the prediction of torque-speed characteristics [3], as well as for the prediction of the cogging torque waveform [4] etc.

Eid and Mouilett [5] proposed an analytical model for predicting the two-dimensional airgap field distribution of internal rotor motors having a cylindrical permanent magnet with uniform diametric magnetization. In addition they and Gu and Gao [6] used the method of separation of variables to analyse the airgap field of a multi-pole permanent magnet motor modelled in rectangular coordinates. Boules [7] also presented a model of a multi-pole permanent magnet motor in rectangular coordinates but used an equivalent magnet pole-arc to account for the fact that in a cylindrical machine the circumferential width of the

magnet generally varies with radius. In [8] Boules formulated a model in polar coordinates which utilized the concept of equivalent current carrying coils to determine the flux density distribution at the stator and rotor surfaces of a permanent magnet motor. Hence, it could account for the flux focusing effect in the magnets, as well as the direction of magnetization, i.e., radial or parallel, on the useful flux per pole and the airgap flux density waveform. Xiong and Nasar [9] analyzed the field in a linear permanent magnet synchronous machine by using the concept of magnetic charge. Gu and Gao [10] and C. Kramer [11] used a "sub-region" method which used a Fourier series expansion in the slot, airgap, and magnet regions, the coefficients of the Fourier series being determined through the application of continuity boundary conditions between the sub-regions. By using a similar technique and a segmented permanent magnet model, Gu and Gao [12] were able to predict the fringing flux distribution along the axial length of a permanent magnet motor. However, existing analytical methods have not accounted effectively for the effects of stator slotting on the airgap field distribution, which in a permanent magnet motor is critical to the prediction of noise and vibration, cogging torque, etc. In addition they have approximated the relative recoil permeability of the permanent magnets, i.e., μ_r , as unity, which can cause an error in the calculation of the flux density of up to around 20% for $\mu_r = 1.2$ [8].

Whilst numerical methods for field computation, such as finite elements, also provide an accurate means of determining the flux density distribution, with due account of saturation etc., they are often time-consuming and do not provide nearly as much insight as analytical solutions into the underlying behavior. Therefore an improved analytical technique is proposed for the calculation of the open-circuit airgap field distribution. It is based on a two-dimensional model in polar coordinates and can cater for both internal and external rotor topologies. It involves the solution of the governing Laplacian/quasi-Poissonian field equations in the airgap/magnet regions without any assumption regarding the relative recoil permeability of the magnets other than the fact that it is constant. However, in this paper the analysis is restricted to slotless motors, a method for modelling the effect of slotting being developed in a subsequent paper. Results from the analysis are

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compared with predictions from corresponding finite element analyses of various radial-field motors.

II. MAGNETIC FIELD PRODUCED BY MAGNETS

A. General Equations In Polar Coordinates

In this section general expressions are derived, in terms of polar coordinates, for the magnetic field produced by the permanent magnets in both external and internal rotor topologies of slotless motors having surface-mounted magnet rotors. The modifying effect of stator slotting is accounted for subsequently in part III [14].

The field vectors $\vec{B}$ and $\vec{H}$ are coupled by

$$\vec{B}_I = \mu_o \vec{H}_I \quad \text{in the airspaces;} \quad (1)$$

$$\vec{B}_{II} = \mu_m \vec{H}_{II} + \mu_o \vec{M} \quad \text{in the permanent magnet} \quad (2)$$

where $\vec{M}$ is the residual magnetization vector, $\mu_m = \mu_o \mu_r$ is the recoil permeability, and μ_r is the relative recoil permeability. For the case of a multi-pole machine equipped with radially magnetized magnets having a linear second-quadrant demagnetization characteristic, $M = B_r/\mu_o$, B_r is the remanence, and the magnetization distribution is shown in Fig. 1.

In terms of the scalar magnetic potential φ

$$\vec{H} = -\nabla\varphi \quad (3)$$

and

$$\nabla^2\varphi_I = 0 \quad \text{in the airspaces} \quad (4)$$

and

$$\nabla^2\varphi_{II} = \frac{\text{div } \vec{M}}{\mu_r} \quad \text{in the magnets.} \quad (5)$$

In rectangular coordinates the magnetization $\vec{M}$ is given by

$$\vec{M} = M_x \vec{i} + M_y \vec{j}$$

where

$$M_x = 0$$

and

$$M_y = M = \sum_{n=1,3,5,\dots}^{\infty} M_n \cos n \frac{\pi}{\tau_p} x \quad (6)$$

and

$$M_n = 2(B_r/\mu_o)\alpha_p \frac{\sin \frac{n\pi\alpha_p}{2}}{\frac{n\pi\alpha_p}{2}} \quad (7)$$

where τ_p is the pole-pitch, and α_p is the magnet pole-arc to pole-pitch ratio. Therefore

$$\text{div } \vec{M} = \frac{\partial M_x}{\partial x} + \frac{\partial M_y}{\partial y} = 0 \quad (8)$$

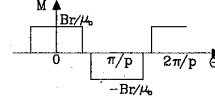


Fig. 1. Radial magnetization (for the case of the magnet pole arc < pole-pitch).

Therefore the scalar magnetic potential in both the airspaces and the permanent magnets is described by the Laplacian equation

$$\frac{\partial^2\varphi}{\partial x^2} + \frac{\partial^2\varphi}{\partial y^2} = 0 \quad (9)$$

where φ is related to the components of the magnetic field strength $\vec{H}$ by (3), giving

$$H_x = -\frac{\partial\varphi}{\partial x}$$

and

$$H_y = -\frac{\partial\varphi}{\partial y} \quad (10)$$

In polar coordinates, however, the magnetization $\vec{M}$ is given by

$$\vec{M} = M_r \vec{r} + M_\theta \vec{\theta}$$

where

$$M_r = \sum_{n=1,3,5,\dots}^{\infty} M_n \cos np\theta$$

and

$$M_\theta = 0 \quad (11)$$

where p is the number of pole-pairs, and the variable “ θ ” is with reference to the center of a magnet pole.

For motors having either an internal or external rotor, the magnetization is assumed to be uniform throughout the cross-section of the magnets, as shown in Fig. 2, and M_n is given by

$$M_n = 2(B_r/\mu_o)\alpha_p \frac{\sin \frac{n\pi\alpha_p}{2}}{\frac{n\pi\alpha_p}{2}} \quad (12)$$

Therefore

$$\text{div } \vec{M} = \frac{M_r}{r} + \frac{\partial M_r}{\partial r} + \frac{1}{r} \frac{\partial M_\theta}{\partial \theta} = \frac{M_r}{r} \quad (13)$$

Therefore, whilst the scalar magnetic potential distribution in the airspace is again governed by Laplace's equation, in the permanent magnet regions it is governed by a quasi-Poissonian equation, i.e.,

$$\frac{\partial^2\varphi_I}{\partial r^2} + \frac{1}{r} \frac{\partial\varphi_I}{\partial r} + \frac{1}{r^2} \frac{\partial^2\varphi_I}{\partial \theta^2} = 0 \quad \text{in the airspaces} \quad (14a)$$

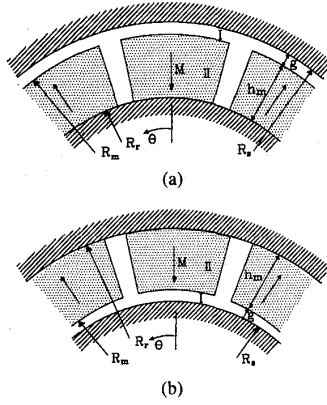


Fig. 2. Motor topologies. a) Internal rotor motor. b) External rotor motor.

and

$$\begin{aligned} \frac{\partial^2 \varphi_{II}}{\partial r^2} + \frac{1}{r} \frac{\partial \varphi_{II}}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \varphi_{II}}{\partial \theta^2} \\ = \frac{M_r}{r\mu_r} \text{ in the permanent magnets} \end{aligned} \quad (14b)$$

where both φ_I and φ_{II} are related to the components of $\vec{H}$ by:

$$\begin{aligned} H_r &= -\frac{\partial \varphi}{\partial r} \\ H_\theta &= -\frac{1}{r} \frac{\partial \varphi}{\partial \theta} \end{aligned} \quad (15)$$

B. General Solutions In Polar Coordinates

For both internal and external rotor motors, such as those shown in Fig. 2, the general solution for the scalar magnetic potential distribution in the airspaces *I* and the magnets *II* has the form

$$\begin{aligned} \varphi(r, \theta) &= \sum_{n'=1}^{\infty} (A_{n'} r^{n'} + B_{n'} r^{-n'}) \\ &\quad \cdot (C_{n'} \cos n'\theta + D_{n'} \sin n'\theta) \\ &\quad + (A_o \ln r + B_o)(C_o \theta + D_o) \end{aligned} \quad (16)$$

which, due to the periodicity of the field, reduces to

$$\varphi(r, \theta) = \sum_{n=1}^{\infty} (A_n r^{np} + B_n r^{-np}) \cos np\theta. \quad (17)$$

In the magnets

$$\nabla^2 \varphi_{II} = \frac{M_r}{r\mu_r} = \frac{1}{r\mu_r} \sum_{n=1,3,5,\dots}^{\infty} M_n \cos np\theta. \quad (18)$$

Hence by assuming a solution of the form

$$\varphi'_{II} = \sum_{n=1,3,5,\dots}^{\infty} C_1 r \cos np\theta \quad (19)$$

and substituting into (18), the constant C_1 is obtained as

$$C_1 = -\frac{1}{(np)^2 - 1} \frac{M_n}{\mu_r}. \quad (20)$$

Therefore

$$\varphi'_{II} = \sum_{n=1,3,5,\dots}^{\infty} \frac{M_n}{\mu_r [1 - (np)^2]} r \cos np\theta. \quad (21)$$

However this solution is not valid for the particular case of $np = 1$. Hence for $np = 1$, by letting $r = e^t$, (18) becomes:

$$\frac{\partial^2 \varphi_{II}}{\partial t^2} + \frac{\partial^2 \varphi_{II}}{\partial \theta^2} = \frac{M_1}{\mu_m} e^t \cos \theta \quad (22)$$

Now assuming

$$\varphi'_{II} = C_2 t e^t \cos \theta \quad (23)$$

and substituting into (22) the constant C_2 is obtained as:

$$C_2 = \frac{1}{2} \frac{M_1}{\mu_r}. \quad (24)$$

Therefore for $np = 1$,

$$\begin{aligned} \varphi'_{II} &= \frac{1}{2} \frac{M_1}{\mu_r} t e^t \cos \theta \\ &= \frac{1}{2} \frac{M_1}{\mu_r} r \ln r \cos \theta. \end{aligned} \quad (25)$$

Therefore the general solutions of (14a) and (14b) are obtained as:

$$\varphi_I(r, \theta) = \sum_{n=1,3,5,\dots}^{\infty} (A_{nI} r^{np} + B_{nI} r^{-np}) \cos np\theta \quad (26)$$

in the airspaces, and

$$\begin{aligned} \varphi_{II}(r, \theta) &= \sum_{n=1,3,5,\dots}^{\infty} (A_{nII} r^{np} + B_{nII} r^{-np}) \cos np\theta \\ &\quad + \sum_{n=1,3,5,\dots}^{\infty} \frac{M_n}{\mu_r [1 - (np)^2]} r \cos np\theta \\ &\quad \text{for } np \neq 1 \end{aligned} \quad (27)$$

or

$$\begin{aligned} \varphi_{II} &= (A_{1II} r + B_{1II} r^{-1}) \cos \theta \\ &\quad + \frac{1}{2} \frac{M_1}{\mu_r} r \ln r \cos \theta \quad \text{for } np = 1 \end{aligned} \quad (28)$$

in the magnets.

The boundary conditions for the internal and external rotor motors of Fig. 2 are defined by:

$$H_{\theta I}(r, \theta) \big|_{r=R_s} = 0 \quad (29)$$

$$H_{\theta II}(r, \theta) \big|_{r=R_r} = 0 \quad (30)$$

$$B_{rI}(r, \theta) \big|_{r=R_m} = B_{rII}(r, \theta) \big|_{r=R_m} \quad (31)$$

$$H_{\theta I}(r, \theta) \big|_{r=R_m} = H_{\theta II}(r, \theta) \big|_{r=R_m} \quad (32)$$

where

$$R_m = R_s - g$$

and

$$R_r = R_s - g - h_m \text{ for an internal rotor motor;}$$

$$R_m = R_s + g$$

and

$$R_r = R_s + g + h_m \text{ for an external rotor motor.} \quad (33)$$

g being the airgap length, and h_m the radial thickness of the magnets.

Hence when $np \neq 1$ the complete solution for the magnetic field components in the airgap/magnet regions, for both internal or external rotor motor topologies, is deduced as:

In the airspaces:

$$B_{rI}(r, \theta) = \sum_{n=1,3,5,\dots}^{\infty} \frac{\mu_o M_n}{\mu_r} \frac{np}{(np)^2 - 1} R_m^{-(np-1)} \cdot \left\{ \frac{(np-1)R_m^{2np} + 2R_r^{np+1}R_m^{np-1} - (np+1)R_r^{2np}}{\frac{\mu_r+1}{\mu_r}[R_s^{2np} - R_r^{2np}] - \frac{\mu_r-1}{\mu_r}[R_m^{2np} - R_s^{2np}(R_r/R_m)^{2np}]} \right\} \cdot [r^{np-1} + R_s^{2np}r^{-(np+1)}] \cos np\theta \quad (34)$$

$$B_{\theta I}(r, \theta) = \sum_{n=1,3,5,\dots}^{\infty} \frac{(-\mu_o M_n)}{\mu_r} \frac{np}{(np)^2 - 1} R_m^{-(np-1)} \cdot \left\{ \frac{(np-1)R_m^{2np} + 2R_r^{np+1}R_m^{np-1} - (np+1)R_r^{2np}}{\frac{\mu_r+1}{\mu_r}[R_s^{2np} - R_r^{2np}] - \frac{\mu_r-1}{\mu_r}[R_m^{2np} - R_s^{2np}(R_r/R_m)^{2np}]} \right\} \cdot [r^{np-1} - R_s^{2np}r^{-(np+1)}] \sin np\theta. \quad (35)$$

In the magnets:

$$B_{rII}(r, \theta) = \sum_{n=1,3,5,\dots}^{\infty} \mu_o M_n \frac{np}{(np)^2 - 1} R_m^{-(np-1)} \cdot \left\{ \frac{\left(np - \frac{1}{\mu_r} \right) R_m^{2np} + \left(1 + \frac{1}{\mu_r} \right) R_r^{np+1} R_m^{np-1} - \left(np + \frac{1}{\mu_r} \right) R_s^{2np} - \left(1 - \frac{1}{\mu_r} \right) (R_r/R_m)^{np+1} R_s^{2np}}{\frac{\mu_r+1}{\mu_r}[R_s^{2np} - R_r^{2np}] - \frac{\mu_r-1}{\mu_r}[R_m^{2np} - R_s^{2np}(R_r/R_m)^{2np}]} \right\} \cdot [r^{np-1} + R_r^{2np}r^{-(np+1)}] \cos np\theta + \sum_{n=1,3,5,\dots}^{\infty} \mu_o M_n \frac{np}{[(np)^2 - 1]} \left(\frac{R_r}{r} \right)^{np+1} \cos np\theta + \sum_{n=1,3,5,\dots}^{\infty} \mu_o M_n \frac{(np)^2}{[(np)^2 - 1]} \cos np\theta \quad (36)$$

$$B_{\theta II}(r, \theta) = \sum_{n=1,3,5,\dots}^{\infty} (-\mu_o M_n) \frac{np}{(np)^2 - 1} R_m^{-(np-1)} \cdot \left\{ \frac{\left(np - \frac{1}{\mu_r} \right) R_m^{2np} + \left(1 + \frac{1}{\mu_r} \right) R_r^{np+1} R_m^{np-1} - \left(np + \frac{1}{\mu_r} \right) R_s^{2np} - \left(1 - \frac{1}{\mu_r} \right) (R_r/R_m)^{np+1} R_s^{2np}}{\frac{\mu_r+1}{\mu_r}[R_s^{2np} - R_r^{2np}] - \frac{\mu_r-1}{\mu_r}[R_m^{2np} - R_s^{2np}(R_r/R_m)^{2np}]} \right\} \cdot [r^{np-1} - R_r^{2np}r^{-(np+1)}] \sin np\theta + \sum_{n=1,3,5,\dots}^{\infty} \mu_o M_n \frac{np}{[(np)^2 - 1]} \left(\frac{R_r}{r} \right)^{np+1} \sin np\theta - \sum_{n=1,3,5,\dots}^{\infty} \mu_o M_n \frac{np}{[(np)^2 - 1]} \sin np\theta. \quad (37)$$

for the particular case when $np = 1$:

In the airspaces:

$$B_{rl}(r, \theta) = \frac{\mu_o M_1}{2\mu_r} \left\{ \frac{\left(\frac{R_m}{R_s}\right)^2 - \left(\frac{R_r}{R_s}\right)^2 + \left(\frac{R_r}{R_s}\right)^2 \ln \left(\frac{R_m}{R_r}\right)^2}{\frac{\mu_r + 1}{\mu_r} \left[1 - \left(\frac{R_r}{R_s}\right)^2\right] - \frac{\mu_r - 1}{\mu_r} \left[\left(\frac{R_m}{R_s}\right)^2 - \left(\frac{R_r}{R_m}\right)^2\right]} \right\} \cdot \left[1 + \left(\frac{R_s}{r}\right)^2\right] \cos \theta \quad (38)$$

$$B_{\theta l}(r, \theta) = \frac{(-\mu_o M_1)}{2\mu_r r} \left\{ \frac{\left(\frac{R_m}{R_s}\right)^2 - \left(\frac{R_r}{R_s}\right)^2 + \left(\frac{R_r}{R_s}\right)^2 \ln \left(\frac{R_m}{R_r}\right)^2}{\frac{\mu_r + 1}{\mu_r} \left[1 - \left(\frac{R_r}{R_s}\right)^2\right] - \frac{\mu_r - 1}{\mu_r} \left[\left(\frac{R_m}{R_s}\right)^2 - \left(\frac{R_r}{R_m}\right)^2\right]} \right\} \cdot \left[1 - \left(\frac{R_s}{r}\right)^2\right] \sin \theta. \quad (39)$$

In the magnets:

$$B_{rII}(r, \theta) = \frac{\mu_o M_1}{2} \left\{ \frac{\left(\frac{R_m}{R_s}\right)^2 - 1 + \ln \left(\frac{R_m}{R_r}\right) \left[\frac{\mu_r + 1}{\mu_r} \left(\frac{R_r}{R_s}\right)^2 - \frac{\mu_r - 1}{\mu_r} \left(\frac{R_r}{R_m}\right)^2\right]}{\frac{\mu_r + 1}{\mu_r} \left[1 - \left(\frac{R_r}{R_s}\right)^2\right] - \frac{\mu_r - 1}{\mu_r} \left[\left(\frac{R_m}{R_s}\right)^2 - \left(\frac{R_r}{R_m}\right)^2\right]} \right\} \cdot \left[1 + \left(\frac{R_r}{r}\right)^2\right] \cos \theta + \frac{\mu_o M_1}{2} \left[1 - \ln \left(\frac{r}{R_m}\right) + \left(\frac{R_r}{r}\right)^2 \ln \left(\frac{R_m}{R_r}\right)\right] \cos \theta \quad (40)$$

$$B_{\theta II}(r, \theta) = \frac{(-\mu_o M_1)}{2} \left\{ \frac{\left(\frac{R_m}{R_s}\right)^2 - 1 + \ln \left(\frac{R_m}{R_r}\right) \left[\frac{\mu_r + 1}{\mu_r} \left(\frac{R_r}{R_s}\right)^2 - \frac{\mu_r - 1}{\mu_r} \left(\frac{R_r}{R_m}\right)^2\right]}{\frac{\mu_r + 1}{\mu_r} \left[1 - \left(\frac{R_r}{R_s}\right)^2\right] - \frac{\mu_r - 1}{\mu_r} \left[\left(\frac{R_m}{R_s}\right)^2 - \left(\frac{R_r}{R_m}\right)^2\right]} \right\} \cdot \left[1 - \left(\frac{R_r}{r}\right)^2\right] \sin \theta + \frac{(-\mu_o M_1)}{2} \left[-\ln \left(\frac{r}{R_m}\right) - \left(\frac{R_r}{r}\right)^2 \ln \left(\frac{R_m}{R_r}\right)\right] \sin \theta. \quad (41)$$

C. Numerical Computation of Field Distribution

For the internal rotor motor shown in Fig. 2(a)

$$R_m = R_s - g$$

and

$$R_r = R_s - g - h_m$$

and

$$R_s > R_m > R_r. \quad (42)$$

Therefore when $np \neq 1$ the expressions for the field vectors B_r and B_θ can be rewritten in the following forms in order to ease their numerical computation:

In the airgap: $R_m < r < R_s$

$$B_{rl}(r, \theta) = \sum_{n=1,3,5,\dots}^{\infty} \frac{\mu_o M_n}{\mu_r} \frac{np}{(np)^2 - 1} \cdot \left\{ \frac{(np - 1) + 2 \left(\frac{R_r}{R_m}\right)^{np+1} - (np + 1) \left(\frac{R_r}{R_m}\right)^{2np}}{\frac{\mu_r + 1}{\mu_r} \left[1 - \left(\frac{R_r}{R_s}\right)^{2np}\right] - \frac{\mu_r - 1}{\mu_r} \left[\left(\frac{R_m}{R_s}\right)^{2np} - \left(\frac{R_r}{R_m}\right)^{2np}\right]} \right\} \cdot \left[\left(\frac{r}{R_s}\right)^{np-1} \left(\frac{R_m}{R_s}\right)^{np+1} + \left(\frac{R_m}{r}\right)^{np+1} \right] \cos np\theta \quad (43)$$

$$B_{\theta I}(r, \theta) = \sum_{n=1,3,5,\dots}^{\infty} \frac{(-\mu_o M_n)}{\mu_r} \frac{np}{(np)^2 - 1} \left\{ \frac{(np-1) + 2 \left(\frac{R_r}{R_m}\right)^{np+1} - (np+1) \left(\frac{R_r}{R_m}\right)^{2np}}{\frac{\mu_r+1}{\mu_r} \left[1 - \left(\frac{R_r}{R_s}\right)^{2np}\right] - \frac{\mu_r-1}{\mu_r} \left[\left(\frac{R_m}{R_s}\right)^{2np} - \left(\frac{R_r}{R_m}\right)^{2np}\right]} \right\} \cdot \left[\left(\frac{r}{R_s}\right)^{np-1} \left(\frac{R_m}{R_s}\right)^{np+1} - \left(\frac{R_m}{r}\right)^{np+1} \right] \sin np\theta \quad (44)$$

In the magnets: $R_r < r < R_m$

$$B_{rII}(r, \theta) = \sum_{n=1,3,5,\dots}^{\infty} \mu_o M_n \frac{np}{(np)^2 - 1} \cdot \left\{ \frac{\left(np - \frac{1}{\mu_r}\right) \left(\frac{R_m}{R_s}\right)^{2np} + \left(1 + \frac{1}{\mu_r}\right) \left(\frac{R_r}{R_m}\right)^{np+1} \left(\frac{R_m}{R_s}\right)^{2np} - \left(np + \frac{1}{\mu_r}\right) - \left(1 - \frac{1}{\mu_r}\right) \left(\frac{R_r}{R_m}\right)^{np+1}}{\frac{\mu_r+1}{\mu_r} \left[1 - \left(\frac{R_r}{R_s}\right)^{2np}\right] - \frac{\mu_r-1}{\mu_r} \left[\left(\frac{R_m}{R_s}\right)^{2np} - \left(\frac{R_r}{R_m}\right)^{2np}\right]} \right\} \cdot \left[\left(\frac{r}{R_m}\right)^{np-1} + \left(\frac{R_r}{R_m}\right)^{np-1} \left(\frac{R_r}{r}\right)^{np+1} \right] \cos np\theta + \sum_{n=1,3,5,\dots}^{\infty} \mu_o M_n \frac{np}{[(np)^2 - 1]} \left(\frac{R_r}{r}\right)^{np+1} \cos np\theta + \sum_{n=1,3,5,\dots}^{\infty} \mu_o M_n \frac{(np)^2}{[(np)^2 - 1]} \cos np\theta \quad (45)$$

$$B_{\theta II}(r, \theta) = \sum_{n=1,3,5,\dots}^{\infty} \frac{(-\mu_o M_n)}{(np)^2 - 1} \cdot \left\{ \frac{\left(np - \frac{1}{\mu_r}\right) \left(\frac{R_m}{R_s}\right)^{2np} + \left(1 + \frac{1}{\mu_r}\right) \left(\frac{R_r}{r}\right)^{np+1} \left(\frac{R_m}{R_s}\right)^{2np} - \left(np + \frac{1}{\mu_r}\right) - \left(1 - \frac{1}{\mu_r}\right) \left(\frac{R_r}{R_m}\right)^{np+1}}{\frac{\mu_r+1}{\mu_r} \left[1 - \left(\frac{R_r}{R_s}\right)^{2np}\right] - \frac{\mu_r-1}{\mu_r} \left[\left(\frac{R_m}{R_s}\right)^{2np} - \left(\frac{R_r}{R_m}\right)^{2np}\right]} \right\} \cdot \left[\left(\frac{r}{R_m}\right)^{np-1} - \left(\frac{R_r}{R_m}\right)^{np-1} \left(\frac{R_r}{r}\right)^{np+1} \right] \sin np\theta + \sum_{n=1,3,5,\dots}^{\infty} \mu_o M_n \frac{np}{[(np)^2 - 1]} \left(\frac{R_r}{r}\right)^{np+1} \sin np\theta - \sum_{n=1,3,5,\dots}^{\infty} \mu_o M_n \frac{np}{[(np)^2 - 1]} \sin np\theta. \quad (46)$$

However, for the external rotor motor shown in Fig. 2(b)

$$R_m = R_s + g$$

and

$$R_s < R_m < R_r. \quad (47)$$

and

$$R_r = R_s + g + h_m$$

Therefore in order to facilitate numerical computation for the case of $np \neq 1$ (34)–(37) are rewritten as:

In the airgap: $R_s < r < R_m$

$$B_{rI}(r, \theta) = \sum_{n=1,3,5,\dots}^{\infty} \frac{(-\mu_o M_n)}{\mu_r} \frac{np}{(np)^2 - 1} \cdot \left\{ \frac{(np-1) \left(\frac{R_m}{R_r}\right)^{2np} + 2 \left(\frac{R_m}{R_r}\right)^{np-1} - (np+1)}{\frac{\mu_r+1}{\mu_r} \left[1 - \left(\frac{R_s}{R_r}\right)^{2np}\right] - \frac{\mu_r-1}{\mu_r} \left[\left(\frac{R_s}{R_m}\right)^{2np} - \left(\frac{R_m}{R_r}\right)^{2np}\right]} \right\} \cdot \left[\left(\frac{r}{R_m}\right)^{np-1} + \left(\frac{R_s}{R_m}\right)^{np-1} \left(\frac{R_s}{r}\right)^{np+1} \right] \cos np\theta \quad (48)$$

$$B_{\theta I}(r, \theta) = \sum_{n=1,3,5,\dots}^{\infty} \frac{\mu_o M_n}{\mu_r} \frac{np}{(np)^2 - 1} \left\{ \frac{(np-1) \left(\frac{R_m}{R_r}\right)^{2np} + 2 \left(\frac{R_m}{R_r}\right)^{np-1} - (np+1)}{\frac{\mu_r+1}{\mu_r} \left[1 - \left(\frac{R_s}{R_r}\right)^{2np}\right] - \frac{\mu_r-1}{\mu_r} \left[\left(\frac{R_s}{R_m}\right)^{2np} - \left(\frac{R_m}{R_r}\right)^{2np}\right]} \right\} \\ \cdot \left[\left(\frac{r}{R_m}\right)^{np-1} - \left(\frac{R_s}{R_m}\right)^{np-1} \left(\frac{R_s}{r}\right)^{np+1} \right] \sin np\theta. \quad (49)$$

In the magnets: $R_m < r < R_r$

$$B_{rII}(r, \theta) = \sum_{n=1,3,5,\dots}^{\infty} (-\mu_o M_n) \frac{np}{(np)^2 - 1} \\ \cdot \left\{ \frac{\left(np - \frac{1}{\mu_r}\right) + \left(1 + \frac{1}{\mu_r}\right) \left(\frac{R_s}{R_m}\right)^{np+1} \left(\frac{R_s}{R_r}\right)^{np-1} - \left(np + \frac{1}{\mu_r}\right) \left(\frac{R_s}{R_m}\right)^{2np} - \left(1 - \frac{1}{\mu_r}\right) \left(\frac{R_m}{R_r}\right)^{np-1}}{\frac{\mu_r+1}{\mu_r} \left[1 - \left(\frac{R_s}{R_r}\right)^{2np}\right] - \frac{\mu_r-1}{\mu_r} \left[\left(\frac{R_s}{R_m}\right)^{2np} - \left(\frac{R_m}{R_r}\right)^{2np}\right]} \right\} \\ \cdot \left[\left(\frac{r}{R_r}\right)^{np-1} \left(\frac{R_m}{R_r}\right)^{np+1} + \left(\frac{R_m}{r}\right)^{np+1} \right] \cos np\theta - \sum_{n=1,3,5,\dots}^{\infty} \mu_o M_n \frac{np}{[(np)^2 - 1]} \left(\frac{r}{R_r}\right)^{np-1} \cos np\theta \\ + \sum_{n=1,3,5,\dots}^{\infty} \mu_o M_n \frac{(np)^2}{[(np)^2 - 1]} \cos np\theta \quad (50)$$

$$B_{\theta II}(r, \theta) = \sum_{n=1,3,5,\dots}^{\infty} \mu_o M_n \frac{np}{(np)^2 - 1} \\ \cdot \left\{ \frac{\left(np - \frac{1}{\mu_r}\right) + \left(1 + \frac{1}{\mu_r}\right) \left(\frac{R_s}{R_m}\right)^{np+1} \left(\frac{R_s}{R_r}\right)^{np-1} - \left(np + \frac{1}{\mu_r}\right) \left(\frac{R_s}{R_m}\right)^{2np} - \left(1 - \frac{1}{\mu_r}\right) \left(\frac{R_m}{R_r}\right)^{np-1}}{\frac{\mu_r+1}{\mu_r} \left[1 - \left(\frac{R_s}{R_r}\right)^{2np}\right] - \frac{\mu_r-1}{\mu_r} \left[\left(\frac{R_s}{R_m}\right)^{2np} - \left(\frac{R_m}{R_r}\right)^{2np}\right]} \right\} \\ \cdot \left[\left(\frac{r}{R_r}\right)^{np-1} \left(\frac{R_m}{R_r}\right)^{np+1} - \left(\frac{R_m}{r}\right)^{np+1} \right] \sin np\theta + \sum_{n=1,3,5,\dots}^{\infty} \mu_o M_n \frac{np}{[(np)^2 - 1]} \left(\frac{r}{R_r}\right)^{np-1} \sin np\theta \\ - \sum_{n=1,3,5,\dots}^{\infty} \mu_o M_n \frac{np}{[(np)^2 - 1]} \sin np\theta. \quad (51)$$

D. Magnetic Field Distribution At The Stator Surface

In the preceding sections general solutions have been derived for the magnetic field distribution in both internal and external rotor motors. However often it is the radial component of the open-circuit flux density distribution at the stator surface which is of most significance, for ex-

ample, in the calculation of the induced emf waveform, the prediction of the cogging torque, and the radial force spectrum.

At the stator surface, i.e., $r = R_s$, the general expression for the radial component of flux density in both internal and external rotor motors becomes

$$B_{rI}(\theta) = \sum_{n=1,3,5,\dots}^{\infty} 2 \frac{\mu_o M_n}{\mu_r} \frac{np}{(np)^2 - 1} \left(\frac{R_s}{R_m}\right)^{np-1} \\ \cdot \left\{ \frac{(np-1)R_m^{2np} + 2R_r^{np+1}R_m^{np-1} - (np+1)R_r^{2np}}{\frac{\mu_r+1}{\mu_r} [R_s^{2np} - R_m^{2np}] - \frac{\mu_r-1}{\mu_r} [R_m^{2np} - R_s^{2np}(R_r/R_m)^{2np}]} \right\} \cos np\theta \quad (52)$$

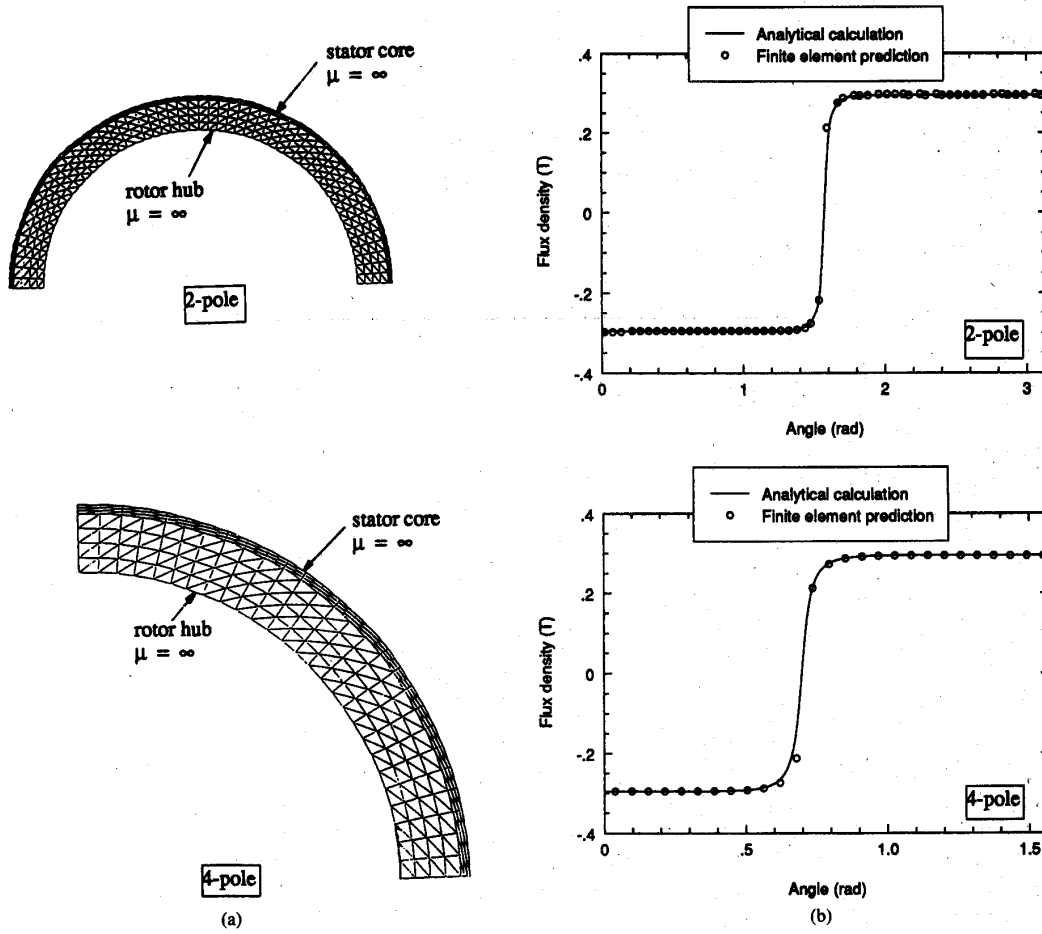


Fig. 3. Comparison of field distribution at the stator surface of internal rotor motors predicted by both finite element and analytical techniques. a) Finite element meshes. b) Field distributions.

when $np \neq 1$, and

$$B_{rl}(\theta) = \frac{\mu_o M_l}{\mu_r} \left\{ \frac{\left(\frac{R_m}{R_s}\right)^2 - \left(\frac{R_r}{R_s}\right)^2 + \left(\frac{R_r}{R_s}\right)^2 \ln \left(\frac{R_m}{R_r}\right)}{\frac{\mu_r + 1}{\mu_r} \left[1 - \left(\frac{R_r}{R_s}\right)^2\right] - \frac{\mu_r - 1}{\mu_r} \left[\left(\frac{R_m}{R_s}\right)^2 - \left(\frac{R_r}{R_m}\right)^2\right]} \right\} \cos \theta \quad (53)$$

when $np = 1$.

For ease of numerical computation the expression for $np \neq 1$ may be written as:
For internal rotor motors ($np \neq 1$)

$$B_{rl}(\theta) = \sum_{n=1,3,5,\dots}^{\infty} 2 \frac{\mu_o M_n}{\mu_r} \frac{np}{(np)^2 - 1} \left(\frac{R_m}{R_s}\right)^{np+1} \left\{ \frac{(np-1) + 2 \left(\frac{R_r}{R_m}\right)^{np+1} - (np+1) \left(\frac{R_r}{R_m}\right)^{2np}}{\frac{\mu_r + 1}{\mu_r} \left[1 - \left(\frac{R_r}{R_s}\right)^{2np}\right] - \frac{\mu_r - 1}{\mu_r} \left[\left(\frac{R_m}{R_s}\right)^{2np} - \left(\frac{R_r}{R_m}\right)^{2np}\right]} \right\} \cos np\theta. \quad (54)$$

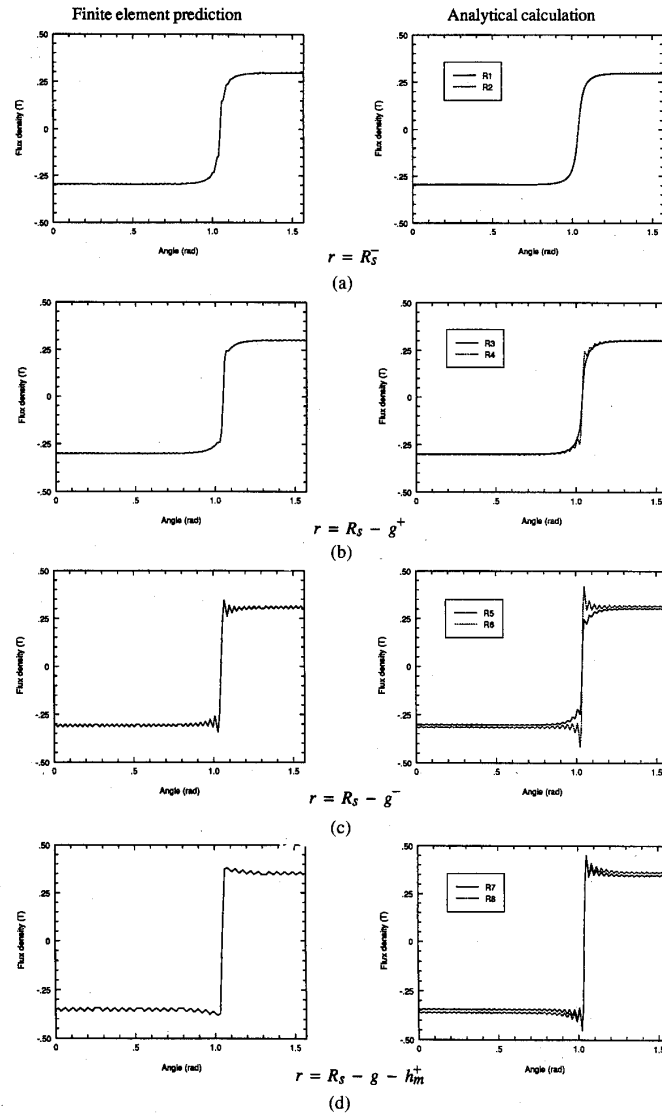


Fig. 4. Variation of radial component of flux density in the 4-pole internal rotor motor at different radii. a) $r = R_s^-$. b) $r = R_s - g^+$. c) $r = R_s - g^-$. d) $r = R_s - g - h_m^+$.

For external rotor motors ($np \neq 1$)

$$B_{rl}(\theta) = \sum_{n=1,3,5,\dots}^{\infty} 2 \frac{(-\mu_o M_n)}{\mu_r} \frac{np}{(np)^2 - 1} \left(\frac{R_s}{R_m} \right)^{np-1} \cdot \left\{ \frac{(np-1) \left(\frac{R_m}{R_r} \right)^{2np} + 2 \left(\frac{R_m}{R_r} \right)^{np-1} - (np+1)}{\frac{\mu_r+1}{\mu_r} \left[1 - \left(\frac{R_s}{R_r} \right)^{2np} \right] - \frac{\mu_r-1}{\mu_r} \left[\left(\frac{R_s}{R_m} \right)^{2np} - \left(\frac{R_m}{R_r} \right)^{2np} \right]} \right\} \cos np\theta. \quad (55)$$

III. COMPARISON WITH THE FINITE ELEMENT CALCULATIONS

The main design parameters of both internal and external rotor topologies of 3-phase brushless dc motors to which the analytical field calculation has been applied are

given in the Appendix. The internal rotor motor is equipped with a sintered ferrite magnet, whilst the external rotor motor has a bonded ferrite magnet, both exhibiting a linear demagnetization characteristic. The analytical predictions have been validated by comparing with

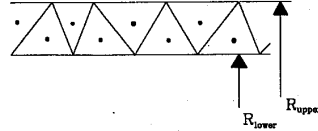


Fig. 5. Effect of finite element discretization.

TABLE I
COMPUTATION RADII AND REGIONS FOR 4-POLE INTERNAL ROTOR MOTOR

Radius	Calculation Radius (m)	Region
$r = R_s^-$	$R_{upper} = R_1$ $R_{lower} = R_2$	airgap
$r = R_s - g^+$	$R_{upper} = R_3$ $R_{lower} = R_4$	airgap
$r = R_s - g^-$	$R_{upper} = R_5$ $R_{lower} = R_6$	magnet
$r = R_s - g - h_m^+$	$R_{upper} = R_7$ $R_{lower} = R_8$	magnet

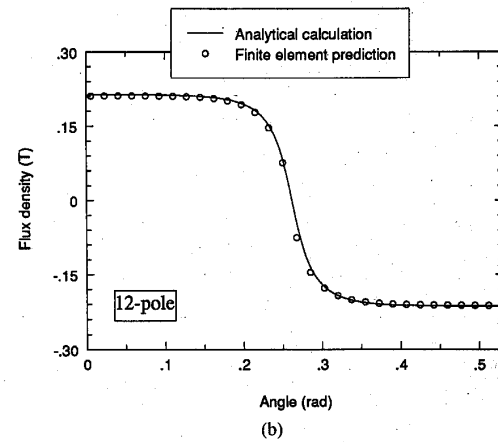
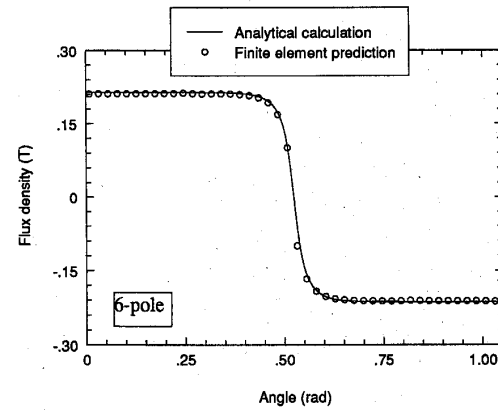
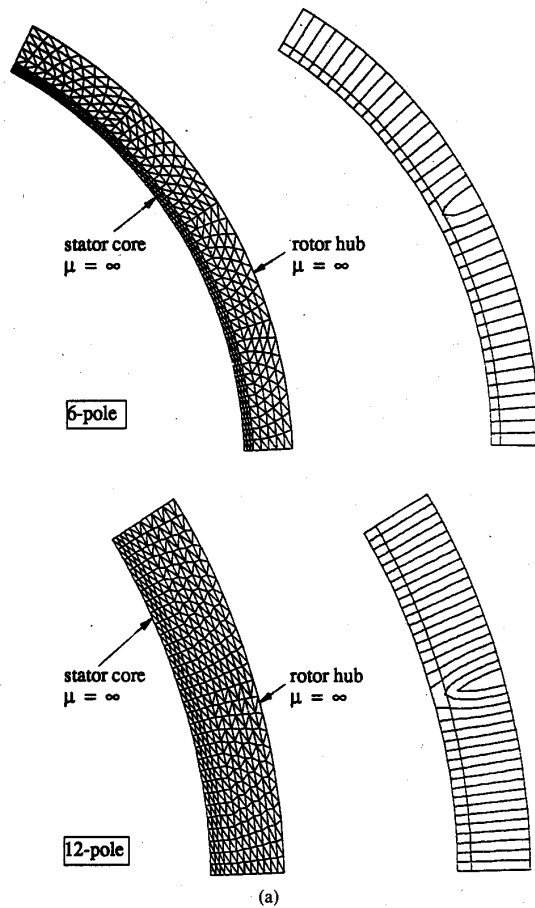


Fig. 6. Finite element mesh and airgap field distribution for external rotor motors. a) Meshes and equi-potential distributions. b) airgap field distributions at the stator surface.

corresponding finite element calculations of the circumferential and radial variations of flux density in passing from the rotor hub, through the airgap/magnet region, to the stator core.

The finite element solutions were obtained by applying a periodic boundary condition at the radial boundaries and imposing the natural Neuman boundary condition at the surface of the stator core and the rotor hub. Fig. 3 com-

compares the open-circuit field distribution at the surface of stator bore for both the 2-pole and 4-pole internal rotor motors, whilst Fig. 4 compares the field distribution at different radii in the magnet/airgap region of the 4-pole motor. Since at each radius the finite element result is plotted from the value of flux density in elements whose node alternate from an upper to a lower radius, R_{upper} and R_{lower} , as shown in Fig. 5, the analytical predictions are given for the two corresponding radii, as shown in Table I. It will be seen that due to the effect of flux leakage the flux density is highest inside the magnets at the circumferential edges of the magnets, whilst outside the magnets, i.e., in the airgap, the flux density is lowest near the edges of the magnets.

It will also be noted that whilst the oscillations which exist in the field distributions plotted from the finite element results are due to the fluctuation of the radial coordinate of the centroid of alternate elements, the amplitude of the oscillation varying with the level of discretisation, the oscillations in the analytical predictions arise through the use of a finite number of terms in the Fourier series, harmonics with an amplitude smaller than 1/500 of the largest harmonic component being neglected. In general, however, the predictions from the analytical technique are in good agreement with the finite element calculations.

Fig. 6 compares analytical and finite element calculations of the airgap flux density distribution at the surface of stator bore for both the 6- and 12-pole external rotor motors, again with the stator slot-openings neglected, and provides further evidence of the utility of the analytical technique.

IV. CONCLUSIONS

A general 2-dimensional analytical method for predicting the open-circuit field distribution in radial-field topologies of permanent magnet motors has been established and validated against finite element calculations. The method can be refined to account for the effect of slotting on the field distribution by the introduction of a "two-dimensional" relative permeance function, as will be described in a companion paper [14] (part III, this issue). Although the method has been applied to brushless dc motors it is equally applicable to brushless permanent magnet ac motors and brushed permanent magnet motors. The analytical model allows the induced emf waveform to be predicted for subsequent use in a dynamic simulation of a brushless dc drive so as to obtain the stator current waveform, thereby allowing the armature reaction field to be determined, as will be described in [13] (part II, this issue). The resultant temporal and spatial dependent magnetic field can then be utilised for predicting the acoustic noise and vibration from a motor, as will be discussed in Part IV, this issue.

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APPENDIX PARAMETERS OF PROTOTYPE MOTORS

TABLE II
PARAMETERS OF PROTOTYPE INTERNAL AND EXTERNAL ROTOR MOTORS

Parameter	Symbol	Internal Rotor	External Rotor	Unit
Pole number	$2p$	2 or 4	6 or 12	—
Slot number	Q_s	12	9	—
Pole-arc/pole-pitch ratio	α_p	1.0	1.0	—
Airgap length	g	0.00075	0.00043	m
Radial thickness of magnet	h_m	0.0045	0.00187	m
Stator radius	R_s	0.02975	0.0207	m
Slot-opening	b_o	0.00205	0.0014	m
Magnet remanence	B_r	0.38	0.25	T
Relative recoil permeability	μ_r	1.05	1.02	—

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